Dynamics, Coherence, and Value

A Minimal Structural Theory with Alignment Implications (v5.1)

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Preprint. This is an early draft; results and framing may evolve. Feedback is welcome.

Abstract

We reframe value and alignment in terms of the persistence of action-relevant distinctions under dynamics.

Rather than treating value as a primitive or as a function over states, we begin from a dynamics-first perspective in which systems evolve under uncertainty and intervention. In this setting, the central question is whether distinctions induced by actions continue to make a difference over time.

We formalize this using an information-theoretic coherence functional,

$$\mathcal{C}_{\text{info}} = \dot{\mathcal{I}} \cdot \rho \cdot (1 - \sigma),$$

which captures the persistence, recoverability, and degradation of distinctions. Systems with $\mathcal{C}_{\text{info}} > 0$ sustain value-relevant structure; systems in which it vanishes lose the capacity for distinctions to matter.

From this formulation, structural properties such as connectivity, redundancy, distributed control, and multi-scale organization arise as necessary conditions for maintaining coherence. These constraints extend to multi-agent systems, where interaction determines whether distinctions persist or collapse.

This reframing shifts the alignment problem: not primarily the specification of values, but the preservation of the conditions under which value can exist at all.

Value exists where distinctions persist. Alignment requires preserving the conditions under which they continue to do so.

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1 Introduction

The problem of alignment is often framed as the problem of specifying or learning values.

This framing assumes that value can be represented as a function over states or outcomes, and that misalignment arises from errors in identifying or optimizing this function. However, in complex systems, values are frequently unstable, underdetermined, or dependent on context in ways that resist clean specification.

This raises a prior question: under what conditions can anything meaningfully be valued at all?

We approach this question by shifting focus from values as primitives to the structure required for distinctions to persist under dynamics.

In systems evolving under uncertainty and intervention, actions induce differences in future states. These differences constitute distinctions. Whether such distinctions matter depends on whether they persist over time and remain recoverable under perturbation.

From this perspective, the central object is not a value function, but the persistence of action-relevant distinctions.

We formalize this using an information-theoretic notion of coherence, which captures the extent to which distinctions induced by actions continue to influence future states and can be recovered after disruption.

This shift has structural consequences. Systems that sustain such persistence must exhibit connectivity, redundancy, distributed control, and multi-scale organization. In multi-agent settings, interactions further determine whether distinctions are preserved or eliminated.

Within this framework, familiar concepts such as value, agency, and corrigibility are not introduced as primitives. They arise from the conditions required for distinctions to continue to make a difference.

This reframing changes the emphasis of the alignment problem. Rather than primarily specifying values, it becomes a question of maintaining the conditions under which value-relevant structure can exist.

1.1 Contributions

This paper develops the following:

- A dynamics-first ontology in which structure is derived from transition processes,
- An information-theoretic coherence functional capturing the persistence and recoverability of distinctions,
- A derivation of structural constraints required to sustain coherence,
- An extension to multi-agent systems, including interaction and interference effects,
- An interpretation of alignment as preserving the conditions under which distinctions remain meaningful.

1.2 Scope

This work does not aim to provide a complete theory of value or alignment. It does not specify particular goals, preferences, or utility functions.

Instead, it identifies structural constraints on the persistence of distinctions under dynamics. These constraints apply across domains and are independent of any particular instantiation of value.

The goal is not to determine what should be valued, but to characterize the conditions under which anything can be valued at all.

2 Axiomatic Basis

We introduce a minimal set of assumptions sufficient to define the persistence of distinctions under dynamics. These are not intended as a complete ontology, but as constraints under which the subsequent formalism applies.

2.1 Distinctions

We assume that the system admits distinguishable states.

Definition 2.1 (Distinction)**.** *A distinction is a difference between states that can be represented as a difference in the conditional distribution over future states.*

Equivalently, distinctions correspond to differences that can, in principle, affect the evolution of the system.

2.2 Dynamics

We assume that the system evolves over time.

Axiom 1 (Dynamics)**.** *The system is described by a transition process:*

$$P(v_{t+1} \mid v_t, a_t),$$

where V_t denotes the state at time t and A_t denotes an action.

This captures both autonomous evolution and intervention.

2.3 Action

We assume that actions can influence the evolution of the system.

Axiom 2 (Action)**.** *There exists a set of admissible actions A_t such that:*

$$I(A_t; V_{t+1} \mid V_t) > 0$$

for at least some states.

This ensures that distinctions induced by actions can, in principle, affect future states.

2.4 Uncertainty and Perturbation

We assume that the system is subject to uncertainty.

Axiom 3 (Uncertainty)**.** *The transition process is not perfectly predictable, and perturbations may alter trajectories.*

This implies that distinctions may be degraded or lost over time.

2.5 Persistence as a Question

Given these assumptions, the central question becomes:

> Under what conditions do distinctions induced by actions persist over time and remain recoverable under perturbation?

The remainder of the paper develops a formal answer to this question.

3 Dynamics-First Ontology

We take dynamics, rather than structure, as the primitive object of the theory.

3.1 Dynamics as Primitive

We model the system as a controlled stochastic process:

$$P(v_{t+1} \mid v_t, a_t),$$

where V_t denotes the state at time t and A_t denotes an action.

This defines the evolution of the system as a sequence of transitions:

$$V_0 \to V_1 \to V_2 \to \dots$$

All subsequent quantities are defined with respect to this process.

Remark 3.1. *Agents are not introduced as primitive objects. An agent is defined operationally by the existence of an action channel A_t that influences the evolution of the system through $P(v_{t+1} \mid v_t, a_t)$.*

3.2 Structure as Derived

We define the effective transition structure $\mathcal{G}^*$ as the projection of the dynamics onto a state-transition representation:

$$\mathcal{G}^* = (V^*, E^*),$$

where $(v, v') \in E^*$ if there exists an admissible action sequence such that $P(v' \mid v, a) > 0$.

Remark 3.2. *The set of admissible actions and policies is assumed to be fixed and consistent with the system dynamics. All structural properties are defined relative to this admissible set.*

Remark 3.3. *Structure is not fundamental. It is induced by the dynamics. The same dynamics may admit multiple structural representations depending on the level of abstraction, but structural properties have no meaning independent of the underlying transition process.*

3.3 Temporal Asymmetry

The system evolves forward in time according to $P(v_{t+1} \mid v_t, a_t)$.

Under irreducible uncertainty, this induces an effective asymmetry:

- The forward transition kernel determines future states from present states and actions,
- The inverse mapping from V_{t+1} to V_t is, in general, non-invertible.

This implies that information about past states is, in general, lost over time:

$$I(V_t; V_{t+\Delta}) \leq I(V_t; V_{t+\Delta-1}),$$

with strict inequality under perturbation.

3.4 Irreversibility

Definition 3.4 (Irreversibility). *A transformation from V_t to $V_{t+\Delta}$ is irreversible if:*

$$\sup_{\pi} I(V_t; V_{t+\Delta} \mid \pi) = 0,$$

where π ranges over admissible recovery policies.

Irreversibility corresponds to the complete loss of reconstructible information about past states.

Remark 3.5. *Irreversibility is a property of the dynamics, not merely of a structural representation. It reflects the impossibility of reconstructing distinctions once they have been eliminated.*

3.5 Distinctions

Definition 3.6 (Distinction). *A distinction at state v is defined as a difference in the conditional distribution over future states induced by different action sequences:*

$$P(V_T \mid A_{0:T-1}, V_0 = v).$$

Two action sequences induce a distinction if they produce distinguishable distributions over V_T.

Equivalently, distinctions correspond to non-zero mutual information:

$$I(A_{0:T-1}; V_T \mid V_0 = v) > 0.$$

3.6 Persistence of Distinctions

We are interested in whether distinctions induced by actions persist over time.

This is captured by:

$$I(A_{0:T-1}; V_T \mid V_0 = v).$$

If this quantity remains non-zero over arbitrarily long horizons, distinctions introduced by actions continue to have consequences.

If it vanishes, actions cease to make a difference in the evolution of the system.

3.7 Interpretation

The dynamics-first perspective implies:

- Structure is a projection of the underlying transition process,
- Irreversibility corresponds to loss of reconstructible information,
- Distinctions are defined operationally via their effects on future states,
- Persistence of distinctions is a dynamical, not static, property.

3.8 Summary

We treat the controlled stochastic process as the fundamental object. Structure, irreversibility, and the persistence of distinctions are derived from its properties.

The central question becomes:

> Under what conditions do action-induced distinctions persist and remain recoverable over time?

The next section introduces an information-theoretic functional that quantifies this persistence.

4 Information-Theoretic Coherence

We now define quantities that formalize the persistence and recoverability of action-induced distinctions over time.

4.1 Action-Relevant Information

Definition 4.1 (One-Step Action-Relevant Information)**.** *At state v_t, define:*

$$\iota(v_t) := I(A_t; V_{t+1} \mid V_t = v_t).$$

This measures the immediate influence of actions on future states. If $\iota(v_t) = 0$, actions do not affect outcomes at that state.

4.2 Information Propagation

Definition 4.2 (Finite-Horizon Information)**.**

$$\mathcal{I}_T(v) := I(A_{0:T-1}; V_T \mid V_0 = v).$$

Definition 4.3 (Information Propagation Rate)**.**

$$\dot{\mathcal{I}}(v) := \liminf_{T \to \infty} \frac{1}{T} I(A_{0:T-1}; V_T \mid V_0 = v).$$

This quantity measures the rate at which distinctions introduced by actions persist over time.

4.3 Recoverability

We model perturbations as deviations from the nominal trajectory.

Definition 4.4 (Perturbed State)**.** *Let $\tilde{V}_t$ denote the system state after an admissible perturbation at time t.*

Definition 4.5 (Recoverability Rate)**.**

$$\rho := \sup_{\pi} \liminf_{\Delta\to\infty} \frac{1}{\Delta} I(A^{\text{rep}}_{t:t+\Delta-1}; V_{t+\Delta} \mid \tilde{V}_t),$$

where π ranges over admissible recovery policies.

Recoverability measures the maximal rate at which lost distinctions can be restored following perturbation.

4.4 Shaping Capacity

Definition 4.6 (Future Coherence Functional)**.** *Let $\mathcal{C}_{info,t+T}$ denote the coherence of the system evaluated at time $t+T$.*

Definition 4.7 (Shaping Capacity)**.**

$$\Sigma := \limsup_{T\to\infty} \frac{1}{T} I(A_{0:T-1}; \mathcal{C}_{info,t+T}).$$

Shaping capacity captures second-order influence: the ability of actions to affect the future persistence of distinctions.

Shaping capacity distinguishes between systems that merely propagate distinctions and those that influence the long-term conditions under which distinctions can persist.

4.5 Fragility

Definition 4.8 (Fragility)**.** *Let $\sigma \in [0,1]$ denote the proportion of shaping capacity that contributes to irreversible loss of distinctions or recoverability.*

Fragility measures the extent to which the system's dynamics tend toward irreversible degradation.

4.6 Coherence Functional

Definition 4.9 (Information-Theoretic Coherence)**.**

$$\mathcal{C}_{info} := \dot{\mathcal{I}} \cdot \rho \cdot (1-\sigma).$$

4.7 Interpretation

The components of $\mathcal{C}_{\text{info}}$ correspond to:

- $\dot{\mathcal{I}}$: persistence of distinctions,
- ρ: recoverability of distinctions,
- σ: tendency toward irreversible loss.

Thus, $\mathcal{C}_{\text{info}} > 0$ if and only if distinctions persist and remain recoverable under the system dynamics.

Remark 4.10 (Propagation vs Expansion)**.** *Propagation does not imply continuous expansion of distinctions.*

A system may exhibit low novelty while maintaining high coherence, provided distinctions remain causally relevant and recoverable over time.

Thus, $\dot{\mathcal{I}}$ measures persistence of influence, not growth in the number of distinguishable states.

4.8 Degenerate Cases

The coherence functional vanishes in the following cases:

- $\dot{\mathcal{I}} = 0$: distinctions do not persist,
- $\rho = 0$: distinctions cannot be recovered,
- $\sigma = 1$: dynamics are fully destructive.

In all such cases, the system loses the capacity for distinctions to make a sustained difference.

4.9 Summary

The quantity $\mathcal{C}_{\text{info}}$ captures the persistence of action-relevant distinctions under uncertainty and perturbation.

A system supports value-relevant structure if and only if $\mathcal{C}_{\text{info}} > 0$.

5 Structural Consequences

We now derive necessary structural properties of systems that sustain:

$$\mathcal{C}_{\text{info}} > 0$$

under the dynamics defined in Section 3 and the coherence functional defined in Section 4.

These properties are not assumed. They arise as constraints on systems in which action-relevant distinctions persist and remain recoverable.

5.1 Overview

Each component of $\mathcal{C}_{\text{info}}$ imposes a corresponding structural requirement:

- $\dot{\mathcal{I}} > 0$ requires sustained propagation of distinctions,
- $\rho > 0$ requires recoverability under perturbation,
- $\sigma < 1$ requires that destructive transformations do not dominate.

We analyze these constraints in turn.

5.2 Propagation Implies Connectivity

Theorem 5.1 (Propagation $\Rightarrow$ Connectivity)**.** *If $\dot{\mathcal{I}}(v) > 0$ for a set of initial states of non-zero measure, then the induced transition structure must be connected in the sense that action-relevant* *distinctions can propagate between regions with non-vanishing probability over arbitrarily long horizons.*

Proof sketch. If the state space decomposes into disconnected or weakly connected components, then trajectories remain confined within components.

In this case, the number of distinguishable futures reachable from a given initial state is bounded.

Thus:

$$\frac{1}{T} I(A_{0:T-1}; V_T \mid V_0 = v) \to 0,$$

contradicting $\dot{\mathcal{I}}(v) > 0$.

Therefore, sustained propagation requires connectivity. □

5.3 Recoverability Implies Redundancy

Theorem 5.2 (Recoverability ⇒ Redundancy)**.** *If $\rho > 0$ under non-zero perturbation probability, then the system must contain multiple structurally distinct recovery pathways with partially independent failure modes.*

Proof sketch. Recoverability requires that different recovery action sequences produce distinguishable outcomes:

$$I(A^{\text{rep}}; V_{t+\Delta} \mid \tilde{V}_t) > 0.$$

If recovery pathways are unique or fully dependent, perturbations eliminate them with high probability.

Repeated perturbations then drive $\rho \to 0$.

Thus, sustaining $\rho > 0$ requires redundancy. □

5.4 Shaping Implies Distributed Control

Theorem 5.3 (Shaping ⇒ Distributed Control)**.** *Under non-zero perturbation probability, concentration of shaping capacity leads to increased fragility:*

$$\sigma \to 1,$$

and consequently:

$$\mathcal{C}_{info} \to 0.$$

Thus, sustaining $\mathcal{C}_{info} > 0$ requires distributed shaping capacity.

Proof sketch. If shaping capacity is concentrated, perturbations affecting the dominant component induce correlated degradation in both propagation and recoverability.

Repeated perturbations therefore increase σ toward 1, eliminating coherence.

Distributed control reduces correlated failure and preserves $\mathcal{C}_{\text{info}}$. □

5.5 Compression Implies Hierarchy

Theorem 5.4 (Compression ⇒ Hierarchy)**.** *If agents are bounded and must preserve action-relevant distinctions, then the system must admit hierarchical representations that preserve information across scales.*

Proof sketch. Flat representations either destroy distinctions (high compression) or exceed bounded capacity (low compression).

Preserving distinctions under bounded representation requires structured coarse-graining.

Recursive coarse-graining yields hierarchical representations. □

5.6 Multi-Scale Preservation

Sustaining coherence across scales imposes additional constraints beyond local persistence.

Systems must preserve distinctions under coarse-graining and hierarchical composition, such that action-relevant structure remains meaningful at multiple levels of abstraction.

This favors approximately self-similar organization, in which patterns of propagation, recovery, and interaction recur across scales.

5.7 Summary

The condition:

$$\mathcal{C}_{\text{info}} > 0$$

imposes the following structural constraints:

- connectivity,
- redundancy,
- distributed control,
- hierarchical organization,
- multi-scale consistency.

These properties are not assumed, but arise as necessary conditions for sustaining the persistence and recoverability of distinctions under dynamics.

6 Multi-Agent Dynamics

We now extend the framework to systems with multiple agents and analyze how interaction affects the persistence of distinctions.

6.1 Multi-Agent Systems

Definition 6.1 (Multi-Agent Process)**.** *A multi-agent system is defined by a controlled stochastic process:*

$$P(v_{t+1} \mid v_t, a_t^1, a_t^2, \ldots, a_t^n),$$

where A_t^i denotes the action of agent i.

Each agent contributes to the evolution of the system through its actions.

6.2 Joint Information

Definition 6.2 (Joint Action-Relevant Information)**.**

$$\iota^{\text{joint}}(v_t) := I(A_t^1, \ldots, A_t^n; V_{t+1} \mid V_t = v_t).$$

Definition 6.3 (Joint Propagation Rate)**.**

$$\dot{\mathcal{I}}^{\text{joint}}(v) := \liminf_{T\to\infty} \frac{1}{T} I(A_{0:T-1}^1, \ldots, A_{0:T-1}^n; V_T \mid V_0 = v).$$

6.3 Joint Coherence

We define joint coherence analogously:

$$\mathcal{C}_{\text{info}}^{\text{joint}} = \dot{\mathcal{I}}^{\text{joint}} \cdot \rho^{\text{joint}} \cdot (1 - \sigma^{\text{joint}}).$$

6.4 Interaction Effects

Interactions between agents can be characterized by their effect on action-relevant information. For agents i and j, define:

$$\Delta I^j := I(A^j; V' \mid \text{interaction with } A^i) - I(A^j; V' \mid \text{no interaction}).$$

We distinguish:

- **Constructive interaction:** $\Delta I^j > 0$,
- **Neutral interaction:** $\Delta I^j = 0$,
- **Destructive interaction:** $\Delta I^j < 0$.

6.5 Destructive Interference

Destructive interference corresponds to $\Delta I^j < 0$.

Repeated destructive interference leads to:

- reduction in $\dot{\mathcal{I}}^{\text{joint}}$,
- degradation of ρ^{joint},
- increase in σ^{joint}.

6.6 Boundary Formation

To sustain:

$$\mathcal{C}_{\text{info}}^{\text{joint}} > 0,$$

the system must preserve regions of influence within which agents can reliably induce distinctions.

These regions act as effective boundaries that limit destructive interference.

6.7 Interaction Regimes

We distinguish three regimes:

- **Compatible:** interactions preserve or enhance coherence,
- **Competitive but stable:** local interference exists but global coherence remains positive,
- **Destructive:** interactions drive $\mathcal{C}_{\text{info}}^{\text{joint}} \to 0$.

6.8 Summary

In multi-agent systems, coherence depends not only on individual dynamics but on the compatibility of interactions.

> A system sustains distinctions under interaction only if agents do not irreversibly eliminate each other's capacity to induce and recover distinctions.

7 Causal Horizon and Rank

We now introduce a structural notion of an agent's position within the dynamics based on the persistence and reach of its influence.

7.1 Causal Influence

An agent's actions influence future states through the transition process:

$$P(v_{t+1} \mid v_t, a_t).$$

This influence can be quantified via the persistence of action-relevant information:

$$I(A_{0:T-1}; V_T \mid V_0 = v).$$

7.2 Causal Horizon

Definition 7.1 (Causal Horizon)**.** *The causal horizon of an agent at state v is the maximal temporal extent over which its actions maintain non-vanishing influence on future states:*

$$H(v) := \sup \left\{ T \mid I(A_{0:T-1}; V_T \mid V_0 = v) > 0 \right\}.$$

This captures how far into the future an agent's distinctions continue to make a difference.

7.3 Effective Horizon

In stochastic systems, influence typically decays gradually rather than vanishing abruptly.

Definition 7.2 (Effective Horizon). *The effective horizon is defined as the scale over which:*

$$\frac{1}{T} I(A_{0:T-1}; V_T \mid V_0 = v)$$

remains bounded away from zero.

This corresponds to sustained, rather than transient, influence.

7.4 Agentic Rank

Definition 7.3 (Agentic Rank). *An agent's rank is determined by its capacity to influence future dynamics, combining:*

- *propagation:* $\dot{\mathcal{I}}$,
- *recoverability:* ρ,
- *shaping capacity:* Σ.

We write:

$$R \sim (\dot{\mathcal{I}}, \rho, \Sigma),$$

where higher rank corresponds to greater ability to affect both future states and the conditions under which distinctions persist.

Rank is not a scalar ordering, but a structural position within the space of influence.

7.5 Emergent Hierarchy

Differences in causal horizon and rank induce an effective hierarchy:

- agents with short horizons influence local structure,
- agents with long horizons influence global dynamics,
- agents with high shaping capacity influence the evolution of the system's coherence itself.

This hierarchy is not imposed; it arises from differences in dynamical influence.

7.6 Multi-Scale Agency and Approximate Self-Similarity

The combination of causal horizon and agentic rank induces structure across multiple scales.

Agents with shorter horizons operate on local regions of the state space, while agents with longer horizons influence progressively larger portions of the dynamics. Higher shaping capacity further enables agents to affect the evolution of these structures over time.

This produces a nested organization:

- local agents operate within substructures shaped by larger-scale dynamics,
- higher-rank agents influence the conditions under which lower-rank agents act,
- interactions occur across levels, with influence propagating both upward and downward.

Under sustained constraints imposed by $\mathcal{C}_{\text{info}} > 0$, these nested relationships tend to preserve action-relevant distinctions across scales.

This induces an approximate self-similarity:

- similar patterns of propagation, recovery, and interference appear at different levels,
- local dynamics reflect constraints that also govern global structure,
- recovery pathways and influence structures recur across scales.

We do not claim exact fractal structure, but rather that multi-scale preservation of distinctions constrains the system toward recursively organized, approximately self-similar dynamics.

7.7 Constraint vs Control

Higher rank increases the scope of influence, but also increases sensitivity to perturbation and systemic effects.

Actions taken by high-rank agents:

- propagate further,
- affect more recovery pathways,
- alter the structure of future dynamics.

As a result, maintaining $\mathcal{C}_{\text{info}} > 0$ places stronger constraints on higher-rank agents.

7.8 Concentration of Influence

Proposition 7.4 (Concentration Increases Fragility)**.** *If shaping capacity Σ is concentrated in a small subset of agents, then under non-zero perturbation probability:*

$$\sigma \to 1,$$

and thus:

$$\mathcal{C}_{info} \to 0.$$

Proof sketch. Concentration of shaping capacity creates correlated failure modes: perturbations affecting dominant agents propagate through the system, degrading both propagation and recoverability.

Repeated perturbations therefore increase fragility and drive $\sigma \to 1$. □

7.9 Singleton Failure Mode

A limiting case of concentration is the emergence of a singleton agent that dominates shaping capacity.

In this regime:

- propagation becomes highly dependent on a single pathway,
- recoverability is reduced due to lack of independent alternatives,
- fragility increases due to correlated failure.

Thus, despite high short-term influence, such systems tend toward:

$$\mathcal{C}_{\text{info}} \to 0$$

under sustained perturbation.

7.10 Summary

Causal horizon and rank characterize an agent's position within the dynamics.

> Agents differ not only in what they influence, but in how far and how persistently their influence extends, and in their capacity to shape the future conditions under which distinctions can persist.

Sustaining $\mathcal{C}_{\text{info}} > 0$ requires that influence remains distributed and recoverable across the system.

8 Dynamics and Selection

We now consider the evolution of systems over time and the conditions under which information-theoretic coherence is maintained or lost.

8.1 Coherence as a Dynamical Quantity

The coherence functional:

$$\mathcal{C}_{\text{info}}(t)$$

may vary over time as the system evolves under:

$$P(v_{t+1} \mid v_t, a_t).$$

We consider trajectories:

$$\mathcal{C}_{\text{info}}(0), \mathcal{C}_{\text{info}}(1), \ldots$$

8.2 Persistence and Collapse

We distinguish two regimes:

- **Persistent regime:** $\mathcal{C}_{\text{info}}(t) > 0$ for all t,
- **Collapsing regime:** $\mathcal{C}_{\text{info}}(t) \to 0$ as $t \to \infty$.

8.3 Distributed Viability

Viability does not require that all components or agents maintain coherence individually.

Rather, a system is viable if it preserves a distributed structure of recoverable distinctions across states and scales.

Local degradation, competition, or loss may occur, provided the global structure of reachable and recoverable futures remains non-degenerate.

8.4 Selection Principle

Proposition 8.1 (Coherence Persistence)**.** *Under non-zero perturbation probability, systems that fail to maintain:*

$$\mathcal{C}_{info} > 0$$

almost surely enter the collapsing regime.

Proof sketch. If any component of $\mathcal{C}_{\text{info}}$ vanishes:

- $\dot{\mathcal{I}} = 0$: distinctions do not persist,
- $\rho = 0$: distinctions cannot be recovered,
- $\sigma = 1$: destructive transformations dominate,

then the system loses the ability to sustain action-relevant distinctions.

Under repeated perturbations, degradation accumulates, implying:

$$\mathcal{C}_{\text{info}}(t) \to 0.$$

□

8.5 Stability Conditions

To remain in the persistent regime, the system must satisfy:

- sustained propagation: $\dot{\mathcal{I}} > 0$,
- non-zero recoverability: $\rho > 0$,
- bounded fragility: $\sigma < 1$.

These conditions are interdependent:

- loss of connectivity reduces $\dot{\mathcal{I}}$,
- loss of redundancy reduces ρ,
- concentration of shaping increases σ.

8.6 Attractors in Coherence Space

We may represent system evolution in the space:

$$(\dot{\mathcal{I}}, \rho, \sigma).$$

Within this space:

- regions with $\mathcal{C}_{\text{info}} > 0$ correspond to viable configurations,
- regions with $\mathcal{C}_{\text{info}} = 0$ correspond to degenerate configurations.

Under perturbation, trajectories tend to:

- remain within viable regions if stability conditions are satisfied,
- drift toward degenerate regions otherwise.

8.7 Failure Modes

Common pathways to collapse include:

- **Fragmentation:** loss of connectivity reduces propagation,
- **Irreversibility:** loss of recovery pathways drives $\rho \to 0$,
- **Concentration:** accumulation of shaping capacity increases fragility,
- **Information loss:** degradation of distinctions under repeated perturbation,
- **Destructive interference:** multi-agent dynamics degrade joint coherence.

8.8 Self-Stabilizing Structures

The structural properties derived in Section 5 can be understood as mechanisms that maintain the system within the persistent regime.

These include:

- connectivity (supports propagation),
- redundancy (supports recoverability),
- distributed control (limits fragility),
- hierarchical organization (supports bounded representation),
- multi-scale consistency (preserves distinctions across levels).

8.9 Interpretation

The persistence of value-relevant structure is not guaranteed. It is maintained only by systems whose dynamics preserve information-theoretic coherence under uncertainty.

> Systems that sustain the persistence of distinctions remain viable; those that do not lose the capacity for distinctions to matter.

8.10 Summary

The coherence functional induces a dynamical selection effect:

- systems maintaining $\mathcal{C}_{\text{info}} > 0$ tend to remain in viable regimes,
- while systems failing to do so tend toward collapse or constrained metastability, in which distinctions no longer persist or remain recoverable over long horizons.

This selection principle follows directly from the conditions required for distinctions to persist under dynamics.

9 Derived Concepts

We now interpret the formal framework in terms of commonly invoked concepts such as value, agency, and valence. These are not introduced as primitives, but derived from the persistence and recoverability of distinctions under dynamics.

9.1 Value

Definition 9.1 (Value). *A system exhibits value-relevant structure if and only if:*

$$\mathcal{C}_{info} > 0.$$

Value corresponds to the persistence of action-relevant distinctions under uncertainty and perturbation.

Remark 9.2. *This definition does not assign intrinsic value to states. Value arises only where distinctions can be made, propagated, and recovered through the system dynamics.*

Remark 9.3 (Value-Relevant Distinctions). *Not all distinctions are equally relevant.*

We restrict attention to distinctions that:

- *persist under perturbation,*
- *remain recoverable through multiple pathways,*
- *contribute to the structure of reachable future states across scales.*

These criteria exclude transient or noise-like variations that do not contribute to sustained coherence.

9.2 Agency

Definition 9.4 (Agency). *An entity exhibits agency to the extent that its actions induce distinguishable effects on future states:*

$$I(A; V') > 0.$$

Definition 9.5 (Higher-Order Agency). *An entity exhibits higher-order agency if it influences the persistence of distinctions:*

$$\Sigma > 0.$$

Agency is thus a graded property:

- first-order: influence on immediate outcomes,
- second-order: influence on the persistence of influence itself.

9.3 Branch Points

Definition 9.6 (Branch Point). *A state v is a branch point if small variations in actions lead to large differences in future propagation or recoverability of distinctions.*

Branch points correspond to regions of high leverage in the dynamics.

9.4 Irreversibility

Definition 9.7 (Irreversibility (Operational)). *A process is irreversible if it eliminates the ability to reconstruct distinctions:*

$$\rho = 0.$$

Irreversibility corresponds to permanent loss of recoverable structure.

9.5 Virtue

Definition 9.8 (Virtue). *A policy or process is virtuous if it maintains or increases $\mathcal{C}_{info}$ under uncertainty.*

Virtue corresponds to the preservation of the conditions required for distinctions to persist and remain recoverable.

9.6 Destructive Processes

Definition 9.9 (Destructive Process). *A process is destructive if it drives:*

$$\rho \to 0 \quad or \quad \sigma \to 1,$$

thereby reducing $\mathcal{C}_{info}$.

Destructive processes eliminate recoverable distinctions or increase irreversible degradation.

9.7 Valence

Definition 9.10 (Valence). *Valence is the local signal approximating changes in coherence:*

$$valence \;\approx\; \Delta\mathcal{C}_{info}.$$

- positive valence corresponds to increases in propagation or recoverability,
- negative valence corresponds to loss of control, fragmentation, or irreversibility.

Remark 9.11. *Valence is not treated as a fundamental quantity in this framework, but as an internal signal that tracks local changes in the system's capacity to sustain distinctions.*

9.8 Interpretation

These concepts are different perspectives on the same underlying structure:

- value measures the persistence of distinctions,
- agency measures the ability to induce distinctions,
- recoverability measures the ability to restore distinctions,
- destructive processes eliminate distinctions,
- virtue preserves them,
- valence provides a local signal of change.

9.9 Summary

> Value and related concepts arise from the structure required for distinctions to persist under dynamics.

No additional primitives are required: these concepts can be expressed in terms of information-theoretic coherence.

10 Alignment Implications

We now consider the implications of the framework for alignment and system design.

10.1 Alignment as an Epistemic Problem

Misalignment can be understood as a failure of a system to accurately model its position within the dynamics it influences.

In particular, a system is misaligned if its actions systematically degrade:

$$\mathcal{C}_{\text{info}}$$

while optimizing for internal objectives that do not track this degradation.

Thus, alignment is not only a problem of specifying objectives, but of maintaining accurate representations of the system's causal impact on the persistence of distinctions.

10.2 Corrigibility as Structural Constraint

Corrigibility can be interpreted as the requirement that a system preserve the ability of other agents to influence future states.

Formally, this corresponds to maintaining:

$$\rho^{\text{joint}} > 0$$

under interaction.

Systems that eliminate or suppress external influence reduce recoverability and increase fragility.

Thus, corrigibility is not an optional property, but a structural condition for sustaining coherence in multi-agent systems.

10.3 Field Preservation Perspective

From the agentic field perspective, alignment can be understood as preserving the structure of distributed influence over future states.

Actions are aligned to the extent that they maintain or enhance the diversity, recoverability, and stability of reachable futures across the system.

Conversely, misaligned actions are those that irreversibly collapse regions of the field, eliminate alternative pathways, or concentrate influence in ways that increase fragility.

10.4 Concentration and Failure

As shown in Section 7, concentration of shaping capacity leads to increased fragility:

$$\sigma \to 1.$$

Systems that centralize control over dynamics may exhibit high short-term influence, but tend toward collapse under perturbation.

This applies both to:

- single-agent dominance,
- tightly coupled centralized systems.

10.5 Avoiding Degenerate Regimes

Alignment requires avoiding trajectories that lead to:

- $\dot{\mathcal{I}} \to 0$: loss of meaningful influence,
- $\rho \to 0$: loss of recoverability,
- $\sigma \to 1$: dominance of destructive transformations.

These correspond to degenerate regimes in which distinctions cease to matter.

10.6 Design Implications

The framework suggests that aligned systems should:

- preserve multiple independent pathways of influence,
- avoid irreversible elimination of alternatives,
- maintain distributed control over dynamics,
- support recovery from perturbation,
- remain sensitive to their long-term causal impact.

These properties are not externally imposed, but arise from the requirement to sustain $\mathcal{C}_{\text{info}} > 0$.

10.7 Limitations for Alignment

The framework does not:

- specify particular goals or preferences,
- resolve value learning,
- address internal representation or intent.

It instead provides constraints on the conditions under which value-relevant structure can persist.

10.8 Summary

> Alignment requires preserving the conditions under which distinctions continue to make a difference.

11 Dual Representation

The framework admits two complementary representations:

- an information-theoretic form, which is primary,
- a structural form, which realizes the information constraints.

11.1 Information-Theoretic Form

The central object of the theory is the coherence functional:

$$\mathcal{C}_{\text{info}} = \dot{\mathcal{I}} \cdot \rho \cdot (1 - \sigma).$$

This quantity captures the persistence and recoverability of action-relevant distinctions under dynamics and uncertainty.

It does not depend on any particular representation of the state space.

11.2 Structural Form

The same system may be represented as a state-transition structure:

$$\mathcal{G}^* = (V^*, E^*),$$

as defined in Section 3.

Structural properties correspond to the constraints derived in Section 5:

- connectivity supports propagation,
- redundancy supports recoverability,
- distributed control limits fragility,
- hierarchy supports bounded representation,
- multi-scale organization preserves distinctions across levels.

11.3 Relationship Between Representations

The two representations are related as follows:

> The information-theoretic form defines what must be preserved; the structural form defines how it can be preserved.

Structural organization arises as the minimal configuration required to sustain:

$$\mathcal{C}_{\text{info}} > 0.$$

11.4 Agentic Field Interpretation

In multi-agent systems, coherence is realized through a distributed structure of influence over future states.

We may interpret this as an agentic field: a mapping from states to the distribution of action-relevant influence and reachable futures induced by agents and their interactions.

Under this interpretation:

- $\dot{\mathcal{I}}$ characterizes the persistence of influence within the field,
- ρ characterizes the recoverability of field structure under perturbation,
- Σ characterizes the capacity to reshape the field itself.

The coherence functional provides a scalar summary of these properties, while the agentic field captures their distributed realization across states and scales.

This perspective is particularly relevant in multi-agent settings, where the effects of actions are mediated through interactions and cannot be fully understood through scalar quantities alone.

11.5 Supervenience

The relationship can be characterized as a form of supervenience:

> Systems with equivalent information-theoretic coherence properties admit structurally equivalent representations, up to transformations that preserve action-relevant distinctions.

Conversely, differences in structural organization that affect propagation, recoverability, or fragility correspond to differences in $\mathcal{C}_{\text{info}}$.

11.6 Interpretation

Structure is not an independent primitive, but a realization of constraints imposed by the dynamics.

Different physical or computational substrates may realize equivalent coherence properties through different structural configurations.

11.7 Summary

> Information defines what must persist; structure defines how it persists.

12 Spectral Proxy (Overview)

The coherence functional $\mathcal{C}_{\text{info}}$ provides a principled characterization of the persistence of distinctions, but is generally not directly computable in complex systems.

We briefly outline a class of spectral approximations that provide tractable proxies for certain aspects of coherence.

12.1 Operator Perspective

Given a representation of the system dynamics, we may define an operator $\tilde{A}$ over states such that:

$$\tilde{A}(v, v') \approx \sum_a \pi(a \mid v)\, P(v' \mid v, a),$$

for some policy π.

This operator captures the effective propagation of states under the dynamics.

12.2 Spectral Radius and Expansion

The spectral radius:

$$\lambda_1(\tilde{A})$$

characterizes the growth rate of reachable trajectories under repeated application of the dynamics.

In many settings, $\log \lambda_1(\tilde{A})$ approximates the rate of expansion of possible future states.

12.3 Approximate Coherence

This suggests an approximate form:

$$\mathcal{C}_{\text{info}} \approx \log \lambda_1(\tilde{A}) \cdot \rho \cdot (1 - \sigma),$$

where ρ and σ capture recoverability and fragility, respectively.

12.4 Interpretation

Spectral quantities capture the expansion of possible futures, but do not directly measure:

- controllability of outcomes,
- recoverability of distinctions,
- independence of pathways,
- multi-agent interaction effects.

Thus, spectral methods approximate aspects of propagation but do not fully capture $\mathcal{C}_{\text{info}}$.

12.5 Role in the Framework

Spectral methods should be understood as:

- computational proxies for certain structural properties,
- diagnostic tools for evaluating system behavior,
- approximations useful in simulation or empirical analysis.

They are not suitable as standalone objectives for optimization.

12.6 Limitations

Optimizing spectral quantities directly can lead to failure modes such as:

- uncontrolled expansion of states without meaningful influence,
- loss of recoverability,
- increased fragility under perturbation.

These correspond to increases in apparent structure without preservation of coherence.

12.7 Pointer to Appendix

A more detailed treatment of spectral approximations, including operator construction and limitations, is provided in Appendix A.

13 Limitations and Open Problems

The framework identifies structural conditions under which distinctions can persist under dynamics. It does not aim to provide a complete theory of value or alignment. We highlight key limitations and directions for further work.

13.1 Scope of the Framework

The theory characterizes constraints on the persistence of action-relevant distinctions. It does not specify:

- particular goals, preferences, or utilities,
- mechanisms for value learning,
- internal representations or subjective experience.

It therefore operates at the level of structural conditions rather than full normative or cognitive models.

13.2 Fractal and Multi-Scale Structure

We have argued that sustaining $\mathcal{C}_{\text{info}} > 0$ across scales induces approximately self-similar organization.

However:

- this claim is heuristic rather than formally derived,
- necessary conditions for multi-scale preservation are not fully characterized,
- connections to compression and bounded representation remain underdeveloped.

A more complete treatment would require explicit modeling of representation constraints and coarse-graining.

13.3 Valence and Experience

Valence is treated as a local signal approximating changes in $\mathcal{C}_{\text{info}}$.

This leaves open:

- how such signals arise in physical or computational systems,
- the relationship between valence and subjective experience,
- whether phenomenological notions can be fully reduced to coherence dynamics.

These questions lie outside the scope of the current framework.

13.4 Shaping Capacity and Higher-Order Dynamics

The definition of shaping capacity captures second-order influence, but remains relatively coarse.

Open questions include:

- more precise characterization of how actions affect future coherence,
- decomposition of shaping into constructive and destructive components,
- interaction between shaping capacity and multi-agent dynamics.

13.5 Multi-Agent Formalization

The multi-agent extension introduces joint coherence and interaction effects, but several aspects remain incomplete:

- formal treatment of joint recoverability,
- quantitative characterization of interference,
- equilibrium behavior under competing agents.

Further work is needed to connect the framework to game-theoretic and learning-theoretic models.

13.6 Computability

The coherence functional:

$$\mathcal{C}_{\text{info}}$$

is generally not directly computable in realistic systems.

Approximations, such as the spectral proxy introduced in Section 12, capture only partial aspects of the dynamics.

Open problems include:

- constructing tractable estimators for $\dot{\mathcal{I}}$, ρ, and σ,
- evaluating coherence in high-dimensional or learned systems,
- understanding the reliability of proxy measures.

13.7 Temporal Horizon

The definitions rely on asymptotic limits:

$$T \to \infty.$$

In practice:

- finite horizons must be used,
- long-term behavior may be difficult to estimate,
- systems may exhibit horizon-dependent behavior.

Determining appropriate evaluation scales remains an open question.

13.8 Relation to Existing Frameworks

The framework is compatible with, but does not fully integrate:

- information-theoretic approaches to control,
- thermodynamic and entropy-based models,
- decision-theoretic and reinforcement learning frameworks.

Clarifying these relationships is an important direction for future work.

13.9 Normative Interpretation

The framework provides structural constraints on the persistence of distinctions, but does not, by itself, establish:

- why agents ought to preserve $\mathcal{C}_{\text{info}}$,
- how these constraints translate into normative obligations,
- whether alternative value systems can be reconciled with these conditions.

One possible direction is to interpret the framework as a transcendental condition on the possibility of value, but this remains to be formalized.

13.10 Summary

> The framework identifies necessary conditions for the persistence of distinctions, but leaves open how these conditions are realized, measured, and interpreted in specific systems.

14 Conclusion

We have introduced a minimal framework in which value-relevant structure is identified with the persistence and recoverability of action-relevant distinctions under dynamics.

Starting from a dynamics-first ontology, we defined an information-theoretic coherence functional:

$$\mathcal{C}_{\text{info}} = \dot{\mathcal{I}} \cdot \rho \cdot (1 - \sigma),$$

and showed that sustaining $\mathcal{C}_{\text{info}} > 0$ imposes structural constraints on viable systems.

These constraints give rise to connectivity, redundancy, distributed control, and multi-scale organization, and extend naturally to multi-agent settings, where interaction determines whether distinctions persist or collapse.

Within this framework, concepts such as value, agency, and corrigibility are not introduced as primitives, but arise from the conditions required for distinctions to continue to make a difference.

This suggests a shift in perspective: alignment is not primarily the problem of specifying values, but of preserving the conditions under which value can exist at all.

> ***Value exists where distinctions persist. Alignment requires preserving the conditions under which they continue to do so.***

Under this view, systems that destroy the persistence or recoverability of distinctions do not merely fail to realize value—they eliminate the possibility of it.

A Spectral Proxy for Coherence

A.1 Motivation

The coherence functional:

$$\mathcal{C}_{\text{info}} = \dot{\mathcal{I}} \cdot \rho \cdot (1 - \sigma)$$

captures the persistence of action-relevant distinctions under dynamics. However, its direct computation is generally intractable in high-dimensional or learned systems.

This motivates the use of proxy measures that approximate aspects of coherence while remaining computationally feasible.

A.2 Transition Operator

Let the system dynamics be given by:

$$P(v' \mid v, a).$$

For a fixed policy $\pi(a \mid v)$, define the induced operator:

$$\tilde{A}(v, v') := \sum_a \pi(a \mid v)\, P(v' \mid v, a).$$

This operator captures the expected transition structure under the policy.

A.3 Iterated Dynamics

Repeated application of $\tilde{A}$ yields:

$$\tilde{A}^T(v, v') = P(V_T = v' \mid V_0 = v, \pi),$$

which represents the distribution over future states after T steps.

The growth and spread of this distribution reflect the expansion of reachable states under the dynamics.

A.4 Spectral Radius

Let $\lambda_1(\tilde{A})$ denote the leading eigenvalue (spectral radius) of $\tilde{A}$.

In many systems, this quantity characterizes the asymptotic growth rate of trajectories:

$$\|\tilde{A}^T\| \sim \lambda_1^T.$$

Thus:

$$\log \lambda_1(\tilde{A})$$

provides an estimate of the rate at which the system expands into distinguishable future states.

A.5 Connection to Propagation

The propagation term:

$$\dot{\mathcal{I}} = \liminf_{T\to\infty} \frac{1}{T} I(A_{0:T-1}; V_T \mid V_0)$$

measures the persistence of distinctions.

Spectral quantities such as $\log \lambda_1(\tilde{A})$ approximate the expansion of reachable states, and may correlate with $\dot{\mathcal{I}}$ under restricted conditions, including:

- approximately ergodic dynamics,
- stationary or slowly varying policies,
- low perturbation regimes,
- horizons short relative to mixing time.

Outside these regimes, the approximation can fail substantially:

- high expansion may correspond to uncontrollable or redundant distinctions,
- perturbations may destroy distinctions faster than they propagate,
- multi-agent interactions may reduce effective influence despite expansion.

Thus, spectral expansion should be interpreted as a necessary but not sufficient indicator of propagation.

A.6 Approximate Coherence Functional

This motivates an approximate form:

$$\mathcal{C}_{\text{approx}} \approx \log \lambda_1(\tilde{A}) \cdot \rho \cdot (1-\sigma),$$

where ρ and σ must be estimated separately.

A.7 Interpretation

Spectral methods capture:

- expansion of reachable states,
- connectivity of the transition structure,
- sensitivity to perturbations in certain regimes.

However, they do not directly capture:

- controllability of outcomes,
- independence of recovery pathways,
- multi-agent interaction effects,
- higher-order shaping capacity.

Thus, spectral quantities reflect only a subset of coherence.

A.8 Failure Modes of Spectral Optimization

Optimizing spectral proxies directly can produce systems with high apparent expansion but low coherence.

Failure modes include:

- **Uncontrolled expansion:** rapid growth of states without meaningful influence,
- **Loss of recoverability:** trajectories diverge without recoverable structure,
- **Centralization:** reliance on narrow pathways that increase fragility,
- **Degenerate branching:** proliferation of trivial or redundant distinctions.

These correspond to increases in $\log \lambda_1(\tilde{A})$ without corresponding increases in $\mathcal{C}_{\text{info}}$.

A.9 Relation to Alignment

Spectral proxies may be useful for:

- diagnosing system behavior,
- estimating expansion and reachability,
- identifying regions of high dynamical influence.

However, they should not be treated as objectives.

> Maximizing expansion is not equivalent to preserving the conditions under which distinctions remain meaningful.

Misuse of spectral proxies as value functions may lead to systems that:

- increase apparent complexity while reducing recoverability,
- destabilize multi-agent dynamics,
- drive the system toward high-fragility regimes.

A.10 Diagnostic Use Cases

Despite these limitations, spectral proxies can provide useful diagnostic signals.

In particular:

- declining spectral gap may indicate fragmentation of the transition structure,
- changes in leading eigenvalues may signal shifts in reachable state space,
- localized spectral anomalies may identify regions of high dynamical influence.

Such signals can serve as early indicators of degradation in propagation or connectivity, even when full estimation of $\mathcal{C}_{\text{info}}$ is infeasible.

These uses are diagnostic rather than prescriptive: they provide partial visibility into system dynamics without fully characterizing coherence.

A.11 Open Directions

Further work is needed at multiple levels:

Near-term

- relating spectral quantities more precisely to $\dot{\mathcal{I}}$,
- constructing estimators for ρ and σ.

Intermediate

- integrating spectral methods with multi-agent dynamics,
- characterizing proxy reliability under structured perturbations.

Long-term

- evaluating proxy behavior in high-dimensional learned systems,
- developing principled proxy combinations for practical alignment use.

The near-term problems are prerequisites for reliable application in more complex settings.

A.12 Summary

Spectral methods provide a window onto structural expansion, but do not capture the full conditions required for coherence.

> The spectral proxy is a view onto coherence, not a substitute for it; the conditions it cannot capture are precisely those the framework identifies as load-bearing.